Ordering Fractions and Mixed Numbers

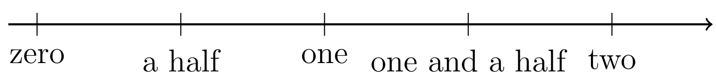
Rewrite into equivalent forms, compare a pair with benchmarks or a common denominator, then order a whole set.

The numbers below arrive in different forms — proper fractions, fractions greater than one, and mixed numbers. Before you can order them you usually have to put them into *one* form. Choose the quickest route each time:

- **Benchmark** — decide first whether each number is below a half, between a half and one, or above one. That alone settles many comparisons.
- **Same numerator or same denominator** — with matching tops, the smaller bottom is the bigger number; with matching bottoms, compare the tops.
- **Common denominator** — when the benchmarks tie, rewrite both numbers over one denominator and compare the numerators.

Show the rewriting you used, then write the answer on the answer line. Use $<$, $>$, or $=$ where a comparison is asked for.

Reference: the benchmark number line. No problem values are shown here — use the numbers given in each problem.



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1. Rewrite $\frac{3}{4}$ as an equivalent fraction with denominator 12.

Answer: ________________

2. Write the mixed number $2\frac{1}{3}$ as a fraction greater than one.

Answer: ________________

3. Which is greater, $\frac{3}{8}$ or $\frac{1}{2}$? Write the two fractions with the correct symbol between them.

Answer: ________________

4. Put these three numbers in order from *least to greatest*:

$$\frac{1}{2}, \qquad \frac{1}{4}, \qquad \frac{3}{8}$$

Answer: ________________

5. Which is greater, $\frac{7}{8}$ or $\frac{5}{6}$? Write the two fractions with the correct symbol between them.
Hint: think about how far each one sits below one.

Answer: ________________

6. Write $\frac{11}{4}$ as a mixed number.

Answer: __________________

7. Put these three numbers in order from *greatest to least*:

$$2\tfrac{1}{2}, \qquad \tfrac{9}{4}, \qquad 2\tfrac{3}{8}$$

Answer: __________________

8. Which is greater, $\frac{5}{9}$ or $\frac{4}{7}$? Write the two fractions with the correct symbol between them.
Careful: the benchmark test does not settle this one.

Answer: __________________

9. Write $\frac{5}{6}$ as an equivalent fraction with denominator 30.

Answer: ________________

10. Put these four numbers in order from *least to greatest*:

$$\frac{3}{5}, \quad \frac{1}{2}, \quad \frac{7}{10}, \quad \frac{13}{20}$$

Answer: ________________

11. Which is greater, $1\frac{3}{4}$ or $\frac{5}{3}$? Write the two numbers with the correct symbol between them.

Answer: ________________

12. Challenge. These four numbers are written in three different forms. Put them in order from *least to greatest*, and show the one common form you converted them all into.

$$\tfrac{5}{3}, \qquad 1\tfrac{1}{2}, \qquad \tfrac{11}{8}, \qquad 1\tfrac{5}{6}$$

Answer: ____________________